D2IO: Diffusion-Based Deep Inertial Odometry for Pedestrian Indoor Localization

Yingying Wang, Shaoting Qiu, Yulong Huang, *Senior Member, IEEE*, and Weisong Wen, *Member, IEEE*

***Abstract*—Indoor localization has been in demand for decades, but no standard solution exists like outdoor global positioning system (GPS). Different from radio frequency-based solutions that require dedicated infrastructure and are susceptible to environmental disturbances, inertial sensors are an ideal choice for indoor localization due to their robustness to environmental variations and low power consumption. Although deep learning has improved the accuracy of open-loop inertial odometry, cumulative errors still pose a significant challenge. In this article, we propose D2IO, an inertial odometry framework that integrates the diffusion model with a transformer architecture. Specifically, the inertial signals are first transformed into a gravity-aligned coordinate frame, followed by the diffusion probabilistic model to augment both the input inertial sequence and output velocity sequence. An inverted transformer framework is then employed to extract spatial and temporal features from each segmented variable. We train the model on a public dataset and evaluate its performance using a 20-h self-collected dataset. The superior results on the challenging evaluation in terms of accuracy and efficiency demonstrate the effectiveness of the proposed D2IO.**

***Index Terms*—Diffusion model, inertial navigation, Internet of Things, velocity estimation.**

I. Introduction

ACCURATE and robust pedestrian indoor localization has been a long-standing aspiration for both academia and industry. Beyond location awareness demand, indoor localization underpins many modern smart services, such as smart homes, healthcare services, and virtual reality applications [1]. Based on whether requiring dedicated infrastructure, pedestrian indoor localization can be categorized into two types: 1) infrastructure-based; and 2) device-free methods [2]. Specifically, the first type of method actively sends signals before receiving echoes from the environment or other sensors, which typically relies on radio frequency-based techniques, such as Bluetooth [3] and ultra-wideband. Moreover, the density of these advanced deployed devices or environmental change significantly affects localization performance [4]. Thus, modalities that depend on interaction between devices or environments are not suitable for ubiquitous indoor localization.

The other device-free localization type relies solely on onboard sensors to measure the ego-motions. Vision and inertial measurement unit (IMU) are two common device-free modalities. However, vision-based localization raises privacy concerns and involves high power consumption. In contrast, IMU is capable of providing a nonintrusive, cost-effective, and environmentally robust solution for relative positioning [5]. Advances in microelectromechanical systems (MEMS) technology have rendered IMU-embedded portable smart devices, such as smartphones and smartwatches, ubiquitous in daily life. The stumbling block to the widespread application of low-cost MEMS-based IMUs is the cumulative position error, which increases at least with the square of the travel time [6]. In recent years, deep learning-based inertial odometry [7], [8], [9] has gained great interest. The powerful capabilities of deep learning facilitate end-to-end velocity regression from noisy and uncalibrated inertial signals. In addition, the generative capacity of deep learning models enables the extraction of periodic and repetitive walking patterns from uncalibrated sensor readings and releases the carrying constraints of IMU embedding devices, i.e., smartphones or smartwatches.

Deep learning-based inertial odometry for pedestrian indoor localization dates back to IONet [10] and RIDI [11], where the smartphone holding manners are categorized into four types. These modes are then processed using independent long short-term memory (LSTM) networks or support vector machines to generate velocity estimates or corrected accelerations. RoNIN [12] removes the constraint of the specific phone-carrying manners and introduces an end-to-end learning framework that transforms 6-D accelerations and angular velocities into planar velocities. The authors also released their dataset, which includes approximately 23 h of publicly available inertial data. Both the end-to-end deep inertial velocity generation framework and the public dataset with unconstrained phone holding modes significantly advance deep learning-based inertial odometry. Since RoNIN, many researchers have been inspired by its residual neural network

Received 13 April 2026; accepted 11 May 2026. This work was supported in part by PolyU Financial Support for Non-PAIR Research Centres under the Project “Advanced Bio-Inspired Spinal System for Humanoid Robots: Design, Manufacturing, and Application in Deep Space Exploration (CE3B).” Paper no. TII-26-3314. *(Corresponding author: Weisong Wen.)*

Yingying Wang, Shaoting Qiu, and Weisong Wen are with the Department of Aeronautical and Aviation Engineering, The Hong Kong Polytechnic University, Hong Kong (e-mail: ying5wang@polyu.edu.hk; shaoting.qiu@connect.polyu.hk; welson.wen@polyu.edu.hk).

Yulong Huang is with the College of Intelligent Systems Science and Engineering, Harbin Engineering University, Harbin 150001, China (e-mail: huangyl@hrbeu.edu.cn).

Digital Object Identifier 10.1109/TII.2026.3693775

(ResNet)-style architecture or have used its dataset to evaluate designed inertial odometry systems.

Existing deep learning techniques have significantly reduced the cumulative errors in inertial odometry. However, long-term accuracy remains challenging, indicating that there is still potential for deep learning to further enhance inertial odometry [13]. A key obstacle to generalization is that real-world inertial sensors exhibit complex, heterogeneous noise beyond simple Gaussian assumptions, including axis misalignment, scale factor instability, and temperature-dependent drift. These characteristics vary across sensor models and deployment environments, leading to both sensor bias and domain shift when models are trained on one dataset and evaluated on another. Recently, diffusion models have become a powerful generative framework, demonstrating remarkable success across computer vision, time-series forecasting, and text generation [14], [15]. Given that inertial signals constitute a form of temporal data and that the diffusion process can implicitly model rich, structured noise distributions, this article investigates how diffusion models can mitigate sensor bias and domain shift to enhance the generalization of inertial odometry. The main contributions are summarized as follows.

1) We design a diffusion-based data augmentation (DA), which incorporates progressive noise-disturbed training samples while maintaining one-time inference efficiency.
2) We introduce dual perturbation on both input latent features and output velocities, with distribution-level and score-matching losses tailored to diffused targets.
3) We propose inverted transformer blocks that independently embed each temporal inertial signal modality.
4) The superior results achieved by the model trained on RoNIN on the 20-h self-collected dataset validate the effectiveness of our system design in mitigating domain shift and handling diverse sensor noises.

II. Related Work

There is little existing research about diffusion model-based inertial signal processing, except in [16] where virtual temporal inertial signals for human activity recognition were generated. In this section, we introduce related work about learning-based inertial navigation and diffusion models separately.

A. Learning-Based Inertial Odometry

TLIO [17] extends the end-to-end inertial odometry system on smartphones, i.e., RoNIN, to head-mounted devices and incorporates an extended Kalman filter (EKF) that uses the integration of inertial measurements as propagation and neural network-regressed velocity as observation. However, the incorporated EKF is not well-suited for smartphone-based inertial odometry, where continuous translation and rotation occur between the device and the pedestrian [18]. IDOL [19] introduces an additional orientation generation stage before the planar velocity regression stage, which requires specially designed hardware to collect ground-truth device orientations and thereby somewhat limits the realism of the inertial signals. Gong et al. [20] enhanced feature construction by utilizing inertial signals captured from both the smartphone and the tightly worn earphone, preprocessing them as the magnitudes of the measured accelerations and angular velocities. CTIN [21] integrates transformer architecture and attention mechanisms into inertial odometry architecture. The inertial measurements at each time step are simultaneously embedded into multivariate temporal tokens, which, however, lacks spatial features capturing among different inertial modalities, i.e., acceleration for translation and angular velocity for rotation. RIO [22] proposes rotation equivariance, which adds the same orientation drift to both input inertial signals and output inertial velocities for DA. SSHNN [23] transforms the many-to-one velocity regression framework into a many-to-many learning system and reduces the inference time by approximately a factor of 10. To ensure high levels of data privacy, federated learning is employed to average the models learned by individual smartphones [24], which requires that the smartphones are homogeneous in their hardware and software configurations. IMUNet [25] incorporates a MobileNet-based [26] noise reduction module, the generated features of which are subtracted in a normal forward feature extraction framework for improved localization efficiency. However, achieving robust generalization across heterogeneous user trajectories is challenging for IMUNet. The 3DIO [8] incorporates additional step detection observations into a TLIO-style regression along with the EKF calibration framework. Although 3DIO and the approach in [27] both employ zero planar pedestrian walking velocity as updated observations, neither approach specifically takes into consideration the significant relative orientation changes between the pedestrian and the device. Specifically, 3DIO only considers scenarios in which the smartphone remains relatively static with respect to the human body, whereas the authors in [27] adopted a quasi-constant orientation assumption. Thus, the performance of both methods can degrade when facing inertial signals containing realistic and complex relative movement between the device and the pedestrian. DeepILS [9] incorporates SSHNN-style convolutional attention mechanisms. However, it is based solely on convolution operations for localization efficiency consideration, resulting in a one-to-one framework rather than the many-to-one structure employed by SSHNN. Feng et al. [28] employed additional magnetic field-based heading estimation to calibrate the learning model regressed inertial velocities, which can be easily affected by changes in the local ferromagnetic surroundings. The frequency of inertial signals has also been explored as model input features using Wavelet encoding [29], where the basis is carefully selected according to walking patterns and may vary among different pedestrians.

B. Diffusion Model

Diffusion models are regarded as a new state-of-the-art (SOTA) family of deep generative models. They exhibit huge potential in various fields, including computer vision, image synthesis, and temporal signal processing [30]. The foundation of diffusion models is to progressively perturb features with random noise, which is known as the "diffusion" process, and then to successively remove noise to generate new samples. The denoising diffusion probabilistic model (DDPM) employs both forward (diffusion) and backward (denoising) processes by two discrete Markov chains [31], with its performance being highly dependent on the noise-adding steps, which is typically on the order of thousands and can significantly affect the overall efficiency. The denoising diffusion implicit model (DDIM) [32] extends DDPM by introducing a non-Markovian deterministic process, allowing for more efficient sampling with fewer diffusion steps while maintaining comparable generative performance. In another approach, the score-based generative model [33] represents both the forward and the reverse processes using continuous stochastic differential equations, thereby accommodating continuous-time dynamics. Furthermore, the conditional diffusion models incorporate additional feature modality into the denoising process instead of relying on pure diffusion process-generated noise and have shown outstanding performance in time-series forecasting and imputation [34]. The latent diffusion model [35] operates the diffusion process in a lower dimensional latent space, enabling more efficient computation and scalability while preserving high-quality generation.

We propose adopting diffusion in the latent space for inertial DA, targeting the generalization challenge caused by sensor bias and domain shift. When deep inertial odometry models are trained on data from one sensor or environment and deployed on another, the mismatch in noise characteristics, such as axis misalignment, scale factor instability, and temperature-dependent drift, leads to significant performance degradation. Existing deep inertial odometry methods typically add random noise to the inertial magnitudes or randomly rotate the inertial vectors [12], [17], [23], which assumes only the basic Allan variance-type noise and therefore fails to capture the structured, nonlinear discrepancies that arise across different sensors and deployment conditions. By applying the diffusion process to both the latent space of inertial signals and the output velocity, our method implicitly learns a richer and more realistic family of noise perturbations, thereby leading the trained model to be robust against the sensor-specific and environment-specific variability encountered at inference time. Oppel and Munz [16] generated synthetic inertial signals providing a specific human activity, which requires thousands of denoising forward propagations. Our diffusion-based DA progressively introduces noise during training while retaining the one-time inference efficiency at test time. Moreover, we introduce an inverted transformer framework to embed inertial signals from each modality along the temporal dimension. The framework leverages the transformer architecture and attention mechanism while preserving the temporal correlations within each inertial signal modality. Extensive experiments validate the effectiveness of our system design.

III. Methodology

A. Preliminary

Given 6-D inertial measurements in the IMU body frame at any measurement time τ, which include 3-axis acceleration $\widetilde{\boldsymbol{a}}_\tau \in \mathbb{R}^3$ and angular velocity $\widetilde{\boldsymbol{\omega}}_\tau \in \mathbb{R}^3$, first step in typical deep inertial odometry is to transform these sensor readings into a pseudo-world coordinate frame. The transformation is achieved through a quasi-equivalent orientation obtained from the integration of angular velocities [6], [23]. The resulting transformed inertial signals are represented as $\boldsymbol{x}_\tau = [\boldsymbol{a}_\tau, \boldsymbol{\omega}_\tau]$. Rotating inertial measurements into the world coordinate frame to serve as network input is a fundamental process in a standard learning-based pedestrian inertial odometry framework. It guides the convergence of the model toward the appropriate regression target, i.e., velocity sequence. The transformed inertial signals are segmented into independent windows of size 200. The generation target is defined as either the instant planar velocity or multiple velocities over a window of size 50 for higher inference efficiency. Consequently, the input and output of the deep learning inference model can be denoted by $\boldsymbol{X}_\tau = [\boldsymbol{x}]_{\tau-200:\tau}{}^T$ and $\boldsymbol{Y}_\tau = [v_x, v_y]_{\tau-50:\tau}$, respectively. For a more detailed illustration about data transformation and segmentation, see [23].

B. Latent Inertial Diffusion Space

In the subsequent discussion, we neglect the time index τ for brevity. Accordingly, our input features and output target are $\boldsymbol{X} \in \mathbb{R}^{6*200}$ and $\boldsymbol{Y} \in \mathbb{R}^{50*2}$, respectively. Adapted to a standard diffusion model architecture, the inertial signals act as conditions for velocity sequence generation. The typical diffusion process is characterized by a fixed Markov chain that gradually transforms initial data $\boldsymbol{Y}$ to random noise $\boldsymbol{Y}_T$

$$q(\boldsymbol{Y}_1, \ldots \boldsymbol{Y}_T|\boldsymbol{Y}) = \prod_{t=1}^{T} q(\boldsymbol{Y}_t|\boldsymbol{Y}_{t-1}) \tag{1}$$

where t denotes the diffusion step index and ranges from 1 to the total number of diffusion steps T. At each step, the distribution $q(\boldsymbol{Y}_t|\boldsymbol{Y}_{t-1})$ incorporates random Gaussian noise into the $\boldsymbol{Y}_{t-1}$ distribution, and is defined as

$$q(\boldsymbol{Y}_t|\boldsymbol{Y}_{t-1}) = \mathcal{N}(\boldsymbol{Y}_t; \sqrt{1-\beta_t}\boldsymbol{Y}_{t-1}, \beta_t \boldsymbol{I}). \tag{2}$$

β_t is a constant uniformly drawn from $(0, \beta_T]$. It can alternatively be set as a deterministic schedule within the interval, such as quadratic or cosine-based schedules. Different choices can modulate the noise variance through the diffusion process, which in turn affects the training quality and evaluation performance. The entire diffusion process gradually transforms the initial $\boldsymbol{Y}$ into random noise $q(\boldsymbol{Y}_T) = \mathcal{N}(0, \beta_T)$ [36]. The addition of noise with various magnitudes to the velocity sequence can make the trained model more robust against different disturbances and result in a more robust velocity sequence. In the reverse (denoising) process, a new fixed Markov chain is defined and is parameterized by the deep model parameter θ to reconstruct $\boldsymbol{Y}$ from random noise

$$p_\theta(\boldsymbol{Y}_0, \ldots \boldsymbol{Y}_{T-1}|\boldsymbol{Y}_T) = \prod_{t=1}^{T} p_\theta(\boldsymbol{Y}_{t-1}|\boldsymbol{Y}_t, \boldsymbol{X}) \tag{3}$$

where the transition function $p_\theta(\boldsymbol{Y}_{t-1}|\boldsymbol{Y}_t, \boldsymbol{X})$ is modeled as a Gaussian distribution $\mathcal{N}(\boldsymbol{Y}_{t-1}; \mu_\theta(\boldsymbol{Y}_t, t), \sigma_\theta(\boldsymbol{Y}_t, t))$. Both the mean μ_θ and the standard deviation σ_θ for $t-1$ step depend on three inputs: 1) the condition $\boldsymbol{X}$; 2) the diffusion step t; and 3) the subsequent noise state $\boldsymbol{Y}_t$, as shown in Fig. 1(a).

Focusing on the reverse process, the denoising unit recursively samples to reconstruct target state from random noise, which is

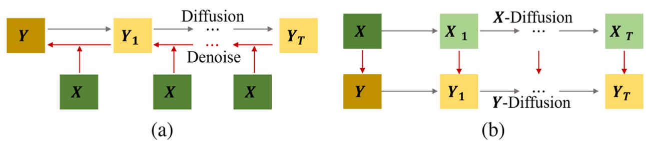


Fig. 1. (a) Typical conditional diffusion model architecture and (b) the proposed diffusion-based inertial samples augmentation. The gray lines indicate the diffusion process, while the red lines are the network model $\boldsymbol{\theta}$ parameterized inference.

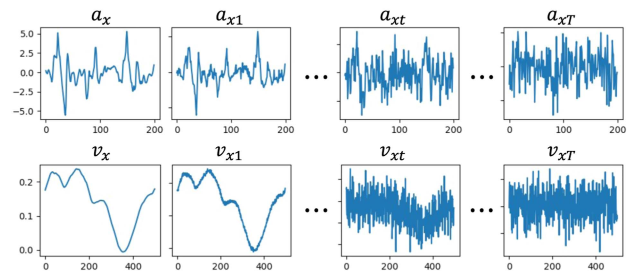


Fig. 2. Diffusion signals visualization. The first and second rows show the diffused states of the raw transformed acceleration and ground-truth velocity in the X-direction, respectively.

time-consuming, especially for complex tasks where the total number of diffusion steps T can reach hundreds to thousands. Instead of applying the diffusion process directly to the raw velocity sequence followed by generating the sequence conditioned on the inertial signal sequence, we propose leveraging only the diffusion process as one of the DA techniques. Specifically, small amounts of Gaussian noise are progressively added to both the transformed input inertial signals $\boldsymbol{X}$ and the output velocity sequence $\boldsymbol{Y}$, thereby creating new training samples, as illustrated in Fig. 1(b). The diffusion-based DA aims to generate diverse training samples that mimic real sensor noise. This approach enhances robustness in inertial odometry by reducing overfitting and improving generalization across varied noise levels with minimal computation consumption, thereby improving localization accuracy and robustness. Specifically, the noise-adding process is inspired by (2). Note that to implement X-diffusion and Y-diffusion processes, we only need to define the number of total diffusion steps T and the progressive noise distribution $\beta_t \in (0, \beta_T]$. Take Y-diffusion as an example, let $\alpha_t = 1 - \beta_t$, we have

$$\begin{aligned}
\boldsymbol{Y}_t &= \sqrt{\alpha_t}\boldsymbol{Y}_{t-1} + \sqrt{1-\alpha_t}\boldsymbol{z}_1 \\
&= \sqrt{\alpha_t}(\sqrt{\alpha_{t-1}}\boldsymbol{Y}_{t-2} + \sqrt{1-\alpha_{t-1}}\boldsymbol{z}_2) + \sqrt{1-\alpha_t}\boldsymbol{z}_1 \\
&= \sqrt{\alpha_t\alpha_{t-1}}\boldsymbol{Y}_{t-2} + \sqrt{\alpha_t(1-\alpha_{t-1})}\boldsymbol{z}_2 + \sqrt{1-\alpha_t}\boldsymbol{z}_1 \\
&= \sqrt{\alpha_t\alpha_{t-1}}\boldsymbol{Y}_{t-2} + \sqrt{1-\alpha_t\alpha_{t-1}}\overline{\boldsymbol{z}}_2 \\
&= \{\ldots \\
&= \sqrt{\overline{\alpha}_t}\boldsymbol{Y} + \sqrt{1-\overline{\alpha}_t}\overline{\boldsymbol{z}}_t.
\end{aligned} \tag{4}$$

$\overline{\alpha}_t$ is the cumulative product up to t step, i.e., $\prod_{i=1}^{t}\alpha_i$, and z_{x} is a normal distribution. $\overline{z}_{\text{x}}$ is also a normal distribution $\mathcal{N}(0, \boldsymbol{I})$ and is derived from the additivity of independent Gaussian distributions. The input feature augmentation, $\boldsymbol{X}$-diffusion, follows an analogous process to $\boldsymbol{Y}$-diffusion. Thus, to achieve DA and construct a latent inertial diffusion space, we only need to specify the total number of diffusion steps and noise parameter β_t. A visualization of $\boldsymbol{X}$-diffusion and $\boldsymbol{Y}$-diffusion is shown in Fig. 2. Furthermore, inspired by diffusion model architectures where the diffusion process is conducted on the decoded feature map rather than on the raw data, the feature diffusion process can also be implemented on the coarse, intermediate feature maps within deep learning models, which will incorporate more structured noise compared to adding noise to the raw data directly.

C. Inverted Inertial Transformer

Transformer architectures have been proven effective in a wide range of applications. In a typical vanilla transformer for time-series processing, the input feature $\boldsymbol{X} \in \mathbb{R}^{6*200}$ is embedded along the temporal direction. In these models, 1-D convolutional kernels slide across time steps, and subsequent feature extractors and attention mechanisms primarily focus on uncovering temporal correlations. However, in our inertial tracking task, the relationship between input features and the generated velocity varies across modalities. For example, accelerations tend to produce higher velocity values, whereas pure angular velocities often correspond to near-zero output. Therefore, it is important to maintain the modality-specific information through the tokenization process in a transformer architecture. Thus, we propose embedding each feature modality independently after the coarse feature extraction stage. The framework of our inverted inertial transformer (iTrans) is illustrated in Fig. 3.

Specifically, the inertial signals are first refined by a combination of a standard convolution operation and a recurrent neural network module to coarse-fine the raw data. The convolution process includes a 1-D convolutional layer, followed by batch normalization, rectified linear unit (ReLU) activation, and a 1-D maxpooling layer. The convolution operation can be mathematically represented as

$$\boldsymbol{C}^T(\boldsymbol{X}) = \text{MaxPool}(\max(\boldsymbol{0}, BN(f_{64}^{7}(\boldsymbol{X})))) \in \mathbb{R}^{64*50}. \tag{5}$$

The input feature is now transformed into 64 different channels, indicating a broader variety of distinct feature patterns and a more compact temporal representation. The subsequent module is a two-layer bidirectional LSTM (BiLSTM) with a hidden size of 128 to further enhance the temporal features. The resulting last hidden state $\boldsymbol{h} \in \mathbb{R}^{50*256}$ is concatenated with the feature map after the convolution operation to form the preprocessed feature map

$$\boldsymbol{F}^T(\boldsymbol{X}) = \text{Concat}([\boldsymbol{C}, \boldsymbol{h}]) \in \mathbb{R}^{50*320}. \tag{6}$$

The preprocessed module merges the local details of the convolution block and the long-term dependencies of the BiLSTM module, providing a concise yet feature-rich representation. After preprocessing, we adopt a transformer architecture for target velocity distribution generation, which includes a feature embedding module, cascaded transformer blocks, and

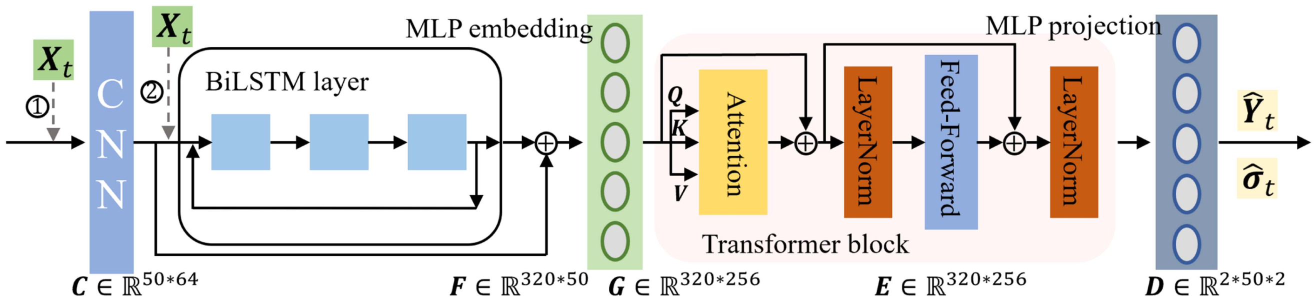


Fig. 3. Framework of the proposed deep inertial odometry system based on diffusion and inverted iTrans blocks. The inertial signals diffusion can be adopted to raw data (①) or after the initial convolution block (②).

multilayer perceptron (MLP)-based projection layers. The main difference is that a typical embedding module employs convolution kernels taking multivariate temporal signals as input and mainly focusing on capturing temporal correlations among all distinct features. In contrast, our inverted MLP layer emphasizes temporal correlations for each feature independently, which can better preserve spatial information extracted by the convolutional neural network (CNN)-LSTM preprocessing module and enrich the temporal representation. Specifically, the embedding layer is designed as a single linear layer that transforms $\boldsymbol{F}$ into a temporal correlation-enriched feature map $\boldsymbol{G} \in \mathbb{R}^{320*256}$.

We adopt a stack of three transformer blocks as the encoder architecture, each of which consists of a self-attention module followed by a feed-forward network. Parameter-free normalization layers are applied both before and after the feed-forward network. The entire transformer preserves the intrinsic dimensions of the feature map. Specifically, the self-attention module treats each distinct temporal feature, denoted by $\boldsymbol{g}_i$ for $i \in \{1, \ldots, 320\}$, as an individual variate and generates the query, key, and value matrices, $\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V} \in \mathbb{R}^{320*256}$, directly from isometric linear transformation of the initial feature map. By employing variatewise dot products, the resulting attention map $\boldsymbol{A} \in \mathbb{R}^{320*320}$ is capable of capturing and accentuating correlations among different variates. The self-attention module can be summarized as

$$\begin{aligned} \boldsymbol{E}_A &= \boldsymbol{A}\boldsymbol{V} + \boldsymbol{G}, \\ &= \text{softmax}\left(\frac{\boldsymbol{Q} \times \boldsymbol{K}^T}{\sqrt{256}}\right)\boldsymbol{V} + \boldsymbol{G}. \end{aligned} \tag{7}$$

The normalization layer is added to increase convergence and enhance training stability. In our framework, it is designed as the standardization for each distinct temporal feature. Take the first normalization layer following the attention module for details. Normalization is performed on the row vector of $\boldsymbol{E}_A$, which is denoted as $\boldsymbol{e}_{Ai}$ for $i \in \{1, \ldots, 320\}$. The normalization layer can be represented as

$$\boldsymbol{E}_{A^L} = (\boldsymbol{e}_{Ai} - \text{Mean}(\boldsymbol{e}_{Ai}))/\sqrt{\text{Var}(\boldsymbol{e}_{Ai})} \tag{8}$$

where $\boldsymbol{E}_{A^L}$ indicates the normalized feature map after the self-attention module.

The feed-forward module serves as the basic building block for extracting features from the embedded representations, which is designed as two pointwise convolutions with an intermediate ReLU activation. As an inverted transformer framework, here we also apply the sliding unit kernel to each independent variate along the temporal dimension instead of performing convolutions across multiple variates that share the same temporal positions. Performing convolutions on each variate independently leverages the assumption that the temporal characteristics of each variate can be treated separately, which has been proven effective on linear forecasters, especially when in cooperation with the inverted MLP embedding [37]. Specifically, the first 1-D convolution encodes 256 variates to 1024 features, and the subsequent convolution reduces them to 256. The final fully connected layer projection module transforms both the variate and temporal dimensions to generate the velocity sequence distribution, i.e., the mean and the corresponding variance.

D. Loss Function Design

Benefiting from the velocity sequence distribution learning objective, we can simultaneously constrain the divergence between two distributions and the relative position derived from the integration within a single forward propagation. The Kullback–Leibler (KL) divergence [38] is employed to quantify the discrepancy between the generative distribution and the diffused target given by (4)

$$\mathcal{L}_{\text{KL}} = D_{\text{KL}}(\hat{\boldsymbol{Y}} || \boldsymbol{Y}_t). \tag{9}$$

$\hat{\boldsymbol{Y}}$ denotes the regressed velocity sequence distribution and is expressed as $\hat{\boldsymbol{Y}} = \hat{\boldsymbol{Y}}_t + \hat{\boldsymbol{\sigma}}_t z$, where z is a standard normal distribution.

We also add a discriminator module after the velocity distribution generator and train it with the denoising score matching (DSM) loss [39]. Using DSM as a final denoising stage has proven effective for suppressing prediction noise and bringing the output closer to the target [40], [41]. Specifically, a residual-based score network predicts the energy of a velocity sequence randomly drawn from the generated distribution $\hat{\boldsymbol{Y}}$. The energy E is expected to scale quadratically with noise magnitude and is formulated as

$$E = (\boldsymbol{w}_1^T \boldsymbol{d} + b_1)(\boldsymbol{w}_2^T \boldsymbol{d} + b_2) + \boldsymbol{w}_3^T (\boldsymbol{d} \odot \boldsymbol{d}) + b_3 \tag{10}$$

where $\boldsymbol{d}$ is the flattened feature map extracted by three cascaded ResBlocks without normalization layers, $\odot$ represents the elementwise multiplication, and $\{\boldsymbol{w}_k, b_k\}_{k=1}^3$ are learnable parameters. The DSM loss is then defined as

$$\mathcal{L}_{\text{DSM}} = \|\boldsymbol{Y}_t - (\tilde{\boldsymbol{Y}} - \nabla_{\tilde{\boldsymbol{Y}}} E)\| \tag{11}$$

where $\tilde{\boldsymbol{Y}}$ is a velocity sequence sampled from the regressed distribution $\hat{\boldsymbol{Y}}$ which serves as a noisy sample. $(\tilde{\boldsymbol{Y}} - \nabla_{\tilde{\boldsymbol{Y}}} E)$ represents the denoised reconstruction, and $\nabla_{\tilde{\boldsymbol{Y}}} E$ is the gradient of the energy with respect to $\tilde{\boldsymbol{Y}}$. Note that the input to the DSM denoising module in the inference stage is the regressed mean velocity value, ensuring deterministic velocity prediction.

During the training stage, instead of directly optimizing the final velocity prediction, i.e., the denoised reconstruction $(\tilde{\boldsymbol{Y}} - \nabla_{\tilde{\boldsymbol{Y}}} E)$, we compute the mean square error (MSE) loss $\mathcal{L}_{\text{MSE}}$ of the randomly sampled velocity $\tilde{\boldsymbol{Y}}$. The sampled velocity sequence includes noise perturbations to the regressed mean value, which forces the network to output highly robust distribution parameters $\hat{\boldsymbol{Y}}$, i.e., an accurate velocity mean with a small variance value. Moreover, the sampled velocity sequences are integrated to construct the relative position sequence formed in a single forward pass of the network. For the sampled velocity sequence $\tilde{\boldsymbol{Y}}$, the corresponding relative position sequence is

$$\overline{\mathbf{p}}_{\tau+\Delta\tau} = 0.01 \cdot \sum_{i=\tau-50}^{\tau+\Delta\tau} \tilde{\boldsymbol{Y}}_i, \quad \text{for } \Delta\tau \in [-50, 0) \tag{12}$$

where the integration constant 0.01 corresponds to the sampling rate of 100 Hz. The 100-Hz sampling rate is obtained by resampling the raw signal measurements, and it has been validated as effective in various typical inertial odometry datasets [10], [17]. The MSE between the position $\overline{\mathbf{p}}_{\tau+\Delta\tau}$ and that derived from the ground-truth velocity is defined as position MSE loss $\mathcal{L}_{\text{PMSE}}$. Note that the ground-truth velocity is the diffusion-based augmented state $\boldsymbol{Y}_t$ during training and the original velocity sequence $\boldsymbol{Y}$ during inference. In other words, the diffusion process is only applied as a DA strategy during the training process. The final loss function is defined as a weighted sum of each loss

$$\mathcal{L} = \mathcal{L}_{\text{MSE}} + \eta\mathcal{L}_{\text{PMSE}} + \gamma\mathcal{L}_{\text{KL}} + \zeta\mathcal{L}_{\text{DSM}} \tag{13}$$

where η, γ, and ζ are the tradeoff weights.

IV. Experiments and Implementation Details

To evaluate the generation capability of the trained models more comprehensively, we train the proposed D2IO system on the public RoNIN dataset while evaluating its performance by the self-collected SHB4 dataset. The collection of the SHB4 dataset follows the configurations of the RoNIN dataset, i.e., the ground-truth positions are collected from the visual-inertial module embedded in a Tango phone, which is attached tightly to the chest of the pedestrian, as shown in Fig. 4. Note that the chest-hung Tango phone is only adopted for the ground-truth pedestrian position collection. The phone can be carried freely in natural modes during inference, i.e., typing, in-pocket, and hand-swinging, as shown in Fig. 4, thereby ensuring user comfort and real-world deployability. Two pedestrians participated in the dataset collection process: one man with a height of 1.83 m and one woman with a height of 1.65 m. Three phones were used in total, as shown in Table I. The entire collection period lasted one month inside an academic building where

(a) (b) (c) (d)

Fig. 4. Selective visualizations of the SHB4 dataset collection. The ASUS Tango phone is hung tightly on the chest, while another phone [Xiaomi 10S in (a) and (c), and Pixel 5 in (b) and (d)] is held for IMU data collection.

TABLE I
Sensors in the SHB4 Dataset

Smartphone	IMU	$\boldsymbol{a}$ noise (ug/$\sqrt{\text{Hz}}$)	$\boldsymbol{\omega}$ noise (mdps/$\sqrt{\text{Hz}}$)
ASUS Tango	InvenSense ICM20602	100	4
Pixel 5	STMicro. LSM6DSR	60	5
Xiaomi 10S	STMicro. LSM6DSO	75	3.8

TABLE II
Details of the Public RoNIN Dataset

	Subjects number	**Trajectories**	**Length (km)**	**Times (h)**
Training	40	72	28.56	11.08
Validation	12	16	6.05	2.27
Test1	30	31	12.91	4.77
Test2	15	32	14.46	5.07

TABLE III
Details of the Self-Collected SHB4 Dataset

	Trajectories	**Length (km)**	**Times (h)**
Train	78	22.28	12.33
Val	14	3.93	1.61
Test	30	10.42	5.79

the furniture layout and temperature were quite consistent with occasional pedestrians passing by. Further details are provided in the Readme file of the online link.[1] The details of these two datasets are provided in Tables II and III, respectively. Note that the dataset split is just for showing the initial split during dataset development. In our experiment, the model is trained on the training, test1, and test2 subsets of the RoNIN dataset, with validation performed on the validation set. We then report evaluations on the entire SHB4 dataset. Testing on the SHB4 dataset is challenging enough for the trained model because of its substantial duration, i.e., about 20 h, and the pronounced domain gap. The pedestrian, device, and location are all new to the model trained on RoNIN.

The number of total diffusion steps is set to 20. A linearly spaced noise scheduler is adopted from 0 to 0.1. The tradeoff weights η, γ, and ζ in the loss function are set to 0.01, 0.01, and 0.03, respectively. We determined the optimal weights through a randomized grid search in preliminary experiments on the RoNIN dataset using the initial dataset split, as shown in Table II. Each weight is evaluated for values in the set $\{0.001, 0.01, 0.1, 1.0\}$. While random search over continuous

[1] Link for SHB4 dataset:. [Online]. Available: https://ieee-dataport.org/documents/shb4-dataset.

distributions is often preferred for its efficiency, we choose to sample from the discrete grid because loss-balancing weights are typically sensitive at the order-of-magnitude level rather than to fine-grained continuous values. Specifically, rather than evaluating the full grid of $4^3 = 64$ points, 30 combinations were randomly selected for computational efficiency reasons. The final weight values were selected based on the best performance of the validation set observed in these experiments. The $\boldsymbol{X}$-diffusion is added after the convolutional preprocessed module according to preliminary experiments. The training is carried out over 150 epochs with a batch size of 128. The learning rate is initialized as 10^{-4} and will decay by a factor of 0.1 if the validation velocity MSE fails to decrease over ten consecutive epochs. The first 5 epochs are warm-up stages with the learning rate linearly increasing from 10^{-5} to 10^{-4}. Dropout with a drop rate of 0.2 is applied after all convolutions within the transformer block and after the hidden layers in the MLP architectures. The network parameters are initialized by Xavier Uniform for multidimensional weights with dim > 1, and scaled to 1 or zero-initialized for other weights and bias vectors, respectively. The entire system is implemented in PyTorch and runs on an NVIDIA RTX 4090D GPU.

V. Results and Analysis

A. Evaluation Metrics and Comparing Methods

To quantitatively assess the performance of the proposed inertial odometry system, we compute the MSE between the generated velocity sequence and the corresponding ground-truth values. Mean absolute error (MAE) is also adopted for the entire average performance evaluations of the velocity sequence and position sequence, respectively. Moreover, the cumulative distribution function (CDF) of location error is plotted to provide a detailed visualization of the localization error distribution. To evaluate the methods more comprehensively, the inference time calculated as the average of 1000 consecutive forward computations with a batch size of 1 is utilized to quantify the model efficiency.

We adopt the ResNet architecture of RoNIN as a baseline method. The SOTA SSHNN, IMUNet, and DeepILS are reproduced on our computer device as comparison methods. We also reimplemented diffusion model-based DDPM and DDIM to demonstrate how pure diffusion models contribute to inertial localization. In the DDPM, the ground-truth velocity sequence, spanning 50 time indexes, is progressively diffused over a maximum of 100 diffusion steps using Gaussian noise with a linearly spaced noise schedule within 10^{-4} to 0.02. Skip-step evaluation with a total of 50 steps is employed during testing. Each denoising step is realized by a conditional diffusion-style UNet, which includes two residual blocks and symmetric residual upsampling paths with skip connections. In the DDIM, the ground-truth velocity sequence is also progressively diffused over a maximum of 100 diffusion steps. The same conditional diffusion-style UNet structure as DDPM is employed to predict the probability distribution of the velocity sequence. However, DDIM adopts a nonlinear noise schedule derived from a deterministic sampling process, allowing for a more efficient and flexible mapping between the noisy and original data.

TABLE IV
Quantitative Localization Error and Inference Time (ms)

Method	MAE_V	MSE_V	MAE_P	50%	75%	Time
RoNIN	0.182	0.072	6.198	8.872	9.006	7.358
SSHNN	0.172	0.065	5.525	8.334	8.669	**0.588**
IMUNet	0.172	**0.064**	5.687	8.700	8.830	10.986
DeepILS	**0.171**	0.064	5.076	7.793	8.025	7.999
DDPM	0.202	0.087	7.559	11.217	11.474	241.643
DDIM	0.196	0.083	6.170	8.555	8.682	237.552
Ours	0.176	0.067	**4.888**	**7.390**	**7.589**	0.889

Bold entities represent the best performance in each row.

B. Experiment Results

The quantitative evaluations of velocity and position errors, along with the velocity inference time over 0.5 s, are summarized in Table IV. The discrepancies of velocity, MAE_V and MSE_V, are reported in m/s, while the remaining position accuracy metrics are presented in m. To regress the velocity sequence over 0.5 s, the RoNIN, IMUNet, and DeepILS methods require ten forward propagation steps, while the remaining methods only run a single time. The proposed method outperforms all compared approaches in position generation, achieving the lowest MAE of regressed positions and reducing the localization error by 5.45% and 5.75% on 50% and 75% of the test data, respectively, compared to the SOTA DeepILS method, which delivers the second-lowest localization error while requiring almost ten times more inference time than the proposed method.

Though SSHNN, IMUNet, and DeepILS achieve lower velocity errors in terms of MAE_V and MSE_V than our method, the integration performance, i.e., position accuracy, is not guaranteed. The reason is that the composition of MSE includes both bias and variance [42]. Bias reflects the average discrepancy of the regressed velocity, which cannot be eliminated and leads to cumulative errors in the integrated position. In contrast, the variance may be reduced during integration. On the test SHB4 dataset, the velocity sequence generated by DeepILS exhibits equivalent bias but higher variance compared to our method, i.e., 0.138 versus 0.132. Thus, though the velocity error of the DeepILS is much lower, its position error remains higher than that of our method. The SSHNN exhibits a shorter inference time than the proposed D2IO method, which may result from its fewer model parameters and reduced floating-point operations (FLOPs). Specifically, SSHNN comprises 2.07-M parameters and a total of 28.57-M FLOPs, whereas D2IO contains 2.99-M parameters and 30.89-M FLOPs. Nevertheless, the D2IO method still meets real-time requirements, achieving an inference time of 0.889 ms for generating a 0.5-s velocity sequence.

The diffusion model-based methods, DDPM and DDIM, provide comparable localization performance to the baseline method, RoNIN. However, their inference time is about 33 times longer than that of RoNIN. In our implementation of DDPM and DDIM, the denoise step is set to 50 to improve efficiency. A higher denoise step T can further enhance the regression accuracy. For example, in the initial DDPM without conditions [31], the T is set to 1000. However, the efficiency would also be significantly reduced. For example, if we set the denoise step the same as the diffusion step without step skip, i.e.,

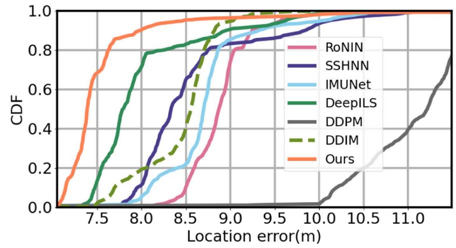


Fig. 5. CDF of location error (m) of our method and comparing methods.

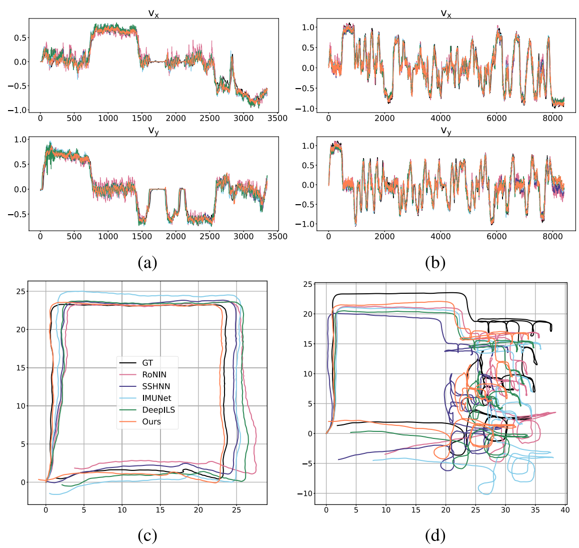


Fig. 6. Selective visualizations of the generated velocity sequence (a) and (b), and the corresponding trajectories obtained by velocity integration (c) and (d). A 0.05 s sampling interval is used in (a) and (b) to match the sampling rate of RoNIN and allow a fair comparison of all competing methods.

$T = 100$, the inference time would double compared to the value reported in Table IV. The high inference time indicates that though diffusion models have the potential to generate accurate temporal velocity sequences, they are inefficient and unsuitable for real-time applications. In contrast, the proposed D2IO method, trained with diffusion-based augmentation while keeping single forward propagation during inference, not only achieves superior positioning performance compared to SOTA methods but also maintains efficiency.

The CDF plot, as shown in Fig. 5, further demonstrates the superior positioning performance of our method compared to all the other methods. The proposed method has the lowest location error across nearly all data segmentation portions, while DeepILS achieves the second-best localization performance. The DDIM demonstrates comparable performance to SSHNN with a smaller maximal location error, though its velocity accuracy is much lower than that of the SSHNN, which further demonstrates the unreliability of instant velocity accuracy in a positioning system. The DDPM performs much worse than other methods, which may be caused by the small maximum diffusion step used in our configuration.

Selective velocity sequence generation results and the corresponding integrated position sequence are presented in Fig. 6. Here, we exclude the inefficient DDPM and DDIM methods

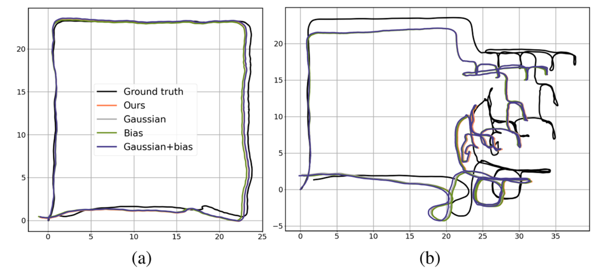


Fig. 7. Selective visualizations of estimated trajectories with various noise disturbance.

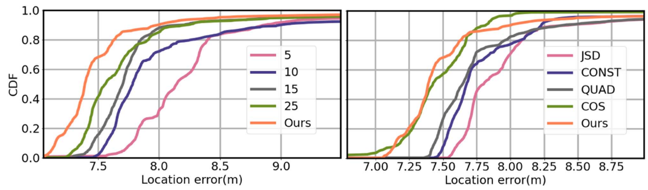
TABLE V
EVALUATION OF D2IO ON VARIOUS DATASETS

Train	Test	MAE_V	MSE_V	MAE_P	50%	75%
RoNIN		0.111	0.030	4.253	6.185	6.355
SHB4		0.114	0.024	4.695	6.179	6.649
RoNIN	SHB4$_t$	0.178	0.074	5.501	8.461	8.764

to better visualize the remaining five methods. The proposed D2IO method consistently outperforms the compared methods, whether in scenarios involving walking with stopping periods [Fig. 6(a) and (c).] or continuous motion [Fig. 6(b) and (d)]. Intuitively, the proposed method demonstrates superior orientation prediction compared to other methods, especially when contrasted with the RoNIN method, which is prone to orientation deviations even for trajectories consisting of simple linear segments with right-angle bends, as illustrated in Fig. 6(a) and (c). The intuition aligns with the quantitative results in Table IV. In summary, both quantitative and qualitative results demonstrate the superior localization performance of our method compared to the competing methods.

To evaluate the robustness of D2IO more comprehensively, bias and additive Gaussian noise are injected into the raw inertial signals. Under a 100-Hz sampling rate, the largest acceleration noise density of 100 $\mathrm{ug}/\sqrt{\mathrm{Hz}}$, as shown in Table I, corresponds to a noise variance of 4.80×10^{-5}, while the largest angular velocity noise density of 5 $\mathrm{mdps}/\sqrt{\mathrm{Hz}}$ corresponding to a noise variance of 3.81×10^{-7}. Accordingly, amplified white Gaussian noise is implemented as $\mathcal{N}(0, 5 \times 10^{-3})$ for acceleration and $\mathcal{N}(0, 4 \times 10^{-5})$ for angular velocity. Fixed bias terms are uniformly drawn from $[-0.2g, 0.2g]$ and $[-2^\circ/s, 2^\circ/s]$ for acceleration and angular velocity, respectively. The Gaussian noise and bias are gradually added to the raw inertial signals. Qualitative visualizations of two representative trajectories under different noise disturbances are shown in Fig. 7. It can be observed that the proposed D2IO performs consistently across different noisy inputs, demonstrating its robustness to sensor noise.

To more comprehensively evaluate our proposed method across different datasets and analyze potential performance gaps, we conducted experiments on the pure RoNIN dataset and the SHB4 dataset separately. The results are shown in Table V. When training and testing on the same dataset, the splitting follows the definitions in Tables II or III. When training on RoNIN and evaluating on SHB4, we show the quantitative results on the test set, i.e., SHB4$_t$ in Table V, for a more direct comparison with the results obtained using only the SHB4 dataset. We can



(a) (b)

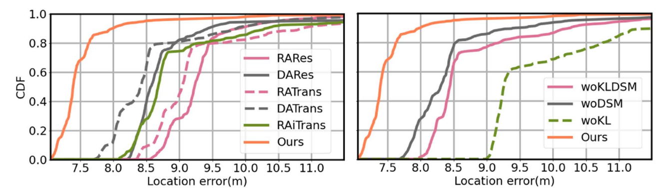
Fig. 8. CDF of location error (m) in the ablation experiments about (a) maximum diffusion step T and (b) noise distribution β_t in the diffusion process.

TABLE VI

QUANTITATIVE RESULTS IN ABLATION EXPERIMENTS ABOUT THE FRAMEWORK

Augment	Model	MAE_V	MSE_V	MAE_P	50%	75%
RA	Res	0.179	0.068	5.981	9.228	9.416
DA	Res	0.177	0.067	5.695	8.566	8.762
RA	Trans	0.177	0.067	5.900	9.039	9.177
DA	Trans	0.177	0.068	5.515	8.411	8.547
RA	iTrans	0.178	0.068	5.563	8.678	9.061
DA	iTrans	**0.176**	**0.067**	**4.888**	**7.390**	**7.589**

Bold entities represent the best performance in each row.



(a) (b)

Fig. 9. CDF of location error (m) in the ablation experiments about (a) framework architecture and (b) loss function design.

observe that the proposed D2IO achieves consistent performance across the three splitting configurations. Within a single dataset, the slightly lower performance on SHB4 compared to RoNIN indicates that SHB4 is somewhat more challenging, e.g., more dynamic interactions between the device and the pedestrian and more irregular trajectories. In the cross-dataset scenario, where the model is trained on RoNIN and evaluated on SHB4$_t$, the performance shows a slight decline compared to that of the single SHB4 dataset evaluation. Such performance degradation in cross-dataset evaluation is common for learning-based methods due to inherent biases, domain shifts between datasets, and the generalization gap of the trained model. The decline is probably due to the domain gap between the two datasets, i.e., different devices, pedestrians, and places. However, the localization performance remains competitive and commendable in this challenging setting. Though the velocity error of the cross-dataset evaluation is noticeably worse than in either of the single-dataset evaluations, the positioning error exhibits quasi-similar accuracy. Overall, the cross-dataset experiments further validate the effectiveness and robustness of our method.

C. Ablation Study

The effectiveness of selecting the diffusion process parameters, i.e., the diffusion step T and noise distribution β_t, is first evaluated. In the visual synthesis task, the diffusion step T is often set to values as high as thousands [43]. However, in the inertial odometry domain, our preliminary experiments indicate that a large T value performs significantly worse than values on the order of tens, which may come from the fact that temporal position information is simpler than the high-dimensional pixel data of images. We show CDF performance across five different diffusion step settings in Fig. 8(a). It can be observed that the employed setting of step 20 achieves the best localization performance across nearly all data percentages. Various noise distributions with the diffusion step set to 20 are also provided in Fig. 8(b), including J-shaped distribution with inverse diffusion step weighting, a constant value of 0.1 (CONST), a quadratic distribution with square-root interpolation (QUAD), and a cosine distribution (COS). It can be observed that the selected linear distribution demonstrates equivalent localization performance to that of COS while outperforming the other three distributions.

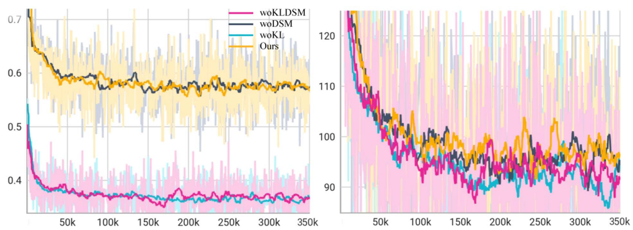
To validate the effectiveness of each of our proposed modules, i.e., the diffusion-based DA and the inverted iTrans block, we conduct ablation experiments by progressively removing each module. Specifically, the diffusion-based augmentation is incorporated into a ResNet-style architecture, which is equivalent to combining the proposed augmentation strategy with RoNIN. The pure inverted iTrans without diffusion-based augmentation is evaluated to investigate how it can enhance the localization performance. Note that when diffusion-based augmentation is excluded, the loss function no longer includes the KL loss and DSM loss. DA is replaced by random rotations and gravity disturbances (RA in Table VI). Specifically, random planar rotations are applied to both the input 6-D inertial signals and the output planar velocities. Random gravity disturbances within 15° [17] are incorporated into the inertial signals to simulate the uncertainties associated with the pseudo-world transformation. When the inverted iTrans module is removed, a ResNet-18 or transformer framework without inverted MLP embedding is employed instead, as shown in Res and Trans in Table VI. Compared to the standard transformer framework, the proposed inverted transformer module has fewer parameters because of its lower embedding dimension. Specifically, iTrans with an embedding dimension of 256 contains 2 396 982 parameters, whereas Trans with an embedding dimension of 320 requires 3 223 486 parameters. On the evaluated device, the single-batch inference time of iTrans is approximately 0.06-ms faster, i.e., 0.810-ms of iTrans versus 0.867 ms of Trans. All ablation experiments are trained on the RoNIN dataset and evaluated on the SHB4 dataset. The quantitative results are provided in Table VI, while the CDF plot is shown in Fig. 9(a). The results indicate that both diffusion-based augmentation and inverted iTrans blocks contribute to improved velocity generation and localization performance. The combination of these two modules makes our D2IO significantly outperform the vanilla ResNet module, achieving a reduction in position MAE by more than 1 m.

We further evaluate the impact of different loss functions on the proposed D2IO framework. Specifically, the KL divergence loss $\mathcal{L}_{\mathrm{KL}}$ and the DSM loss $\mathcal{L}_{\mathrm{DSM}}$ are progressively eroded. The CDF of the loss function module erosion experiments is shown in Fig. 9(b). Unlike the architecture erosion experiments, excluding $\mathcal{L}_{\mathrm{KL}}$ and $\mathcal{L}_{\mathrm{DSM}}$ simultaneously achieves better performance than

TABLE VII
QUANTITATIVE RESULTS IN ABLATION EXPERIMENTS ABOUT THE LOSS FUNCTION

$\mathcal{L}_{KL}$	$\mathcal{L}_{DSM}$	MAE_V	MSE_V	MAE_P	50%	75%
		0.175	**0.067**	5.317	8.441	8.884
✓		0.177	0.069	5.466	8.302	8.478
	✓	0.179	0.068	5.882	9.255	10.455
✓	✓	0.176	**0.067**	**4.888**	**7.390**	**7.589**

Bold entities represent the best performance in each row.



(a) (b)

Fig. 10. MSE loss for(a) velocity $\mathcal{L}_{\text{MSE}}$ and (b) position $\mathcal{L}_{\text{PMSE}}$ over training steps.

excluding $\mathcal{L}_{\text{KL}}$ alone, which can also be observed from the lower MAE of positions listed in Table VII. The largest localization error when excluding $\mathcal{L}_{\text{KL}}$, i.e., woKL in Fig. 9(b), may stem from the fact that DSM in the second stage assumes its input, i.e., the output of the first stage, is with low bias and noise. However, when the KL loss is removed, the output velocity sequence distribution of the first stage may become significantly biased and noisy. Fig. 10 shows the $\mathcal{L}_{\text{MSE}}$ and $\mathcal{L}_{\text{PMSE}}$ over training steps for the four network configurations. Surprisingly, the MSE loss for velocity in D2IO and woDSM, which achieve the top two smallest localization errors, as indicated by Fig. 9(b) and Table VII, converges to a much higher value compared to the other two frameworks. However, the MSE losses for positions, integrated from the generated velocity sequence across the four configurations, do not exhibit significant differences. The ablation experiments on loss function design further suggest that the velocity MSE loss may not directly correlate with localization accuracy and validate the effectiveness of our loss function design.

The ablation experiments further demonstrate the necessity of each component, including the diffusion parameter selections, the diffusion-based DA strategy, the inverted transformer block, and the loss function design. All components enable the proposed D2IO method to track the pedestrian movement using only unconstrained carried IMU devices, outperforming competing methods quantitatively and qualitatively. With the higher localization accuracy, the D2IO method can be integrated with other localization modalities, such as WiFi and GNSS, to develop a more robust, nonintrusive, and seamless indoor–outdoor localization system. On the other hand, diffusion-based DA also has the potential to enhance other tasks, such as human motion prediction and time-series forecasting.

VI. CONCLUSION

This article proposes a hybrid diffusion process applied to both inertial signals and target velocity sequences for DA during training. An inverted transformer architecture is then employed to extract both spatial and temporal features from each preprocessed variate, while an MLP projection generates the final velocity distribution. The loss function is designed to constrain both typical velocity sequence generation and distribution formulation. A rigorous evaluation, including training on a public dataset and testing on a 20-h self-collected dataset, demonstrates the superior performance of the proposed approach compared to SOTA learning-based methods and typical diffusion model-based sequence generation approaches. Extensive ablation experiments further validate the effectiveness of our system design. In future work, we aim to incorporate additional nonintrusive localization modalities, such as ubiquitous WiFi received signal strength, to calibrate the cumulative errors inherent in pure inertial-based localization and further enhance the robustness of pedestrian indoor localization.

REFERENCES

[1] A. Zhang, Z. Min, Y. Wang, and M. Q.-H. Meng, "Towards an accurate augmented-reality-assisted orthopedic surgical robotic system using bidirectional generalized point set registration," in *Proc. IEEE Int. Conf. Intell. Robots Syst.*, 2023, pp. 4600–4607.

[2] R. C. Shit et al., "Ubiquitous localization (ubiloc): A survey and taxonomy on device free localization for smart world," *IEEE Commun. Surv. Tutor.*, vol. 21, no. 4, pp. 3532–3564, Fourth Quarter 2019.

[3] Y. Zhuang, C. Zhang, J. Huai, Y. Li, L. Chen, and R. Chen, "Bluetooth localization technology: Principles, applications, and future trends," *IEEE Internet Things J.*, vol. 9, no. 23, pp. 23506–23524, Dec. 2022.

[4] J. Xiao, Z. Zhou, Y. Yi, and L. M. Ni, "A survey on wireless indoor localization from the device perspective," *ACM Comput. Surv.*, vol. 49, no. 2, pp. 1–31, 2016.

[5] X. Jiang et al., "DDIO-mapping: A fast and robust visual-inertial odometry for low-texture environment challenge," *IEEE Trans. Ind. Inform.*, vol. 20, no. 3, pp. 4418–4428, Mar. 2024.

[6] Y. Wang, H. Cheng, C. Wang, and M. Q.-H. Meng, "Pose-invariant inertial odometry for pedestrian localization," *IEEE Trans. Instrum. Meas.*, vol. 70, 2021, Art. no. 8503512.

[7] C. Chen and X. Pan, "Deep learning for inertial positioning: A survey," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 9, pp. 10506–10523, Sep. 2024.

[8] S. Bai, W. Wen, and C. Shi, "3DIO: Low-drift 3-D deep-inertial odometry for indoor localization using an IMU," *IEEE Internet Things J.*, vol. 12, no. 9, pp. 12711–12722, May 2025.

[9] O. Tariq, M. B. A. Dastagir, M. Bilal, and D. Han, "Deepils: Toward accurate domain-invariant AIoT-enabled inertial localization system," *IEEE Internet Things J.*, vol. 12, no. 11, pp. 17153–17168, Jun. 2025.

[10] C. Chen, X. Lu, A. Markham, and N. Trigoni, "IONet: Learning to cure the curse of drift in inertial odometry," in *Proc. AAAI Conf. Artif. Intell.*, 2018, pp. 6468–6476.

[11] H. Yan, Q. Shan, and Y. Furukawa, "RIDI: Robust IMU double integration," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2018, pp. 621–636.

[12] S. Herath, H. Yan, and Y. Furukawa, "RoNIN: Robust neural inertial navigation in the wild: Benchmark, evaluations, & new methods," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2020, pp. 3146–3152.

[13] Y. Liu et al., "Embodied navigation," *Sci. China Inf. Sci.*, vol. 68, no. 4, 2025, Art. no. 141101.

[14] F.-A. Croitoru, V. Hondru, R. T. Ionescu, and M. Shah, "Diffusion models in vision: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 9, pp. 10850–10869, Sep. 2023.

[15] Y. Huang et al., "Diffusion model-based image editing: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 6, pp. 4409–4437, Jun. 2025.

[16] H. Oppel and M. Munz, "Imudiffusion: A diffusion model for multivariate time series synthetisation for inertial motion capturing systems," 2024, *arXiv:2411.02954*.

[17] W. Liu et al., "TLIO: Tight learned inertial odometry," *IEEE Robot. Autom. Lett.*, vol. 5, no. 4, pp. 5653–5660, Oct. 2020.

[18] Y. Wang, H. Cheng, and M. Q.-H. Meng, "Inertial odometry using hybrid neural network with temporal attention for pedestrian localization," *IEEE Trans. Instrum. Meas.*, vol. 71, 2022, Art. no. 7503610.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

[19] S. Sun, D. Melamed, and K. Kitani, "Idol: Inertial deep orientation-estimation and localization," in *Proc. AAAI Conf. Artif. Intell.*, 2021, pp. 6128–6137.

[20] J. Gong, X. Zhang, Y. Huang, J. Ren, and Y. Zhang, "Robust inertial motion tracking through deep sensor fusion across smart earbuds and smartphone," *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.*, vol. 5, no. 2, pp. 1–26, 2021.

[21] B. Rao, E. Kazemi, Y. Ding, D. M. Shila, F. M. Tucker, and L. Wang, "CTIN: Robust contextual transformer network for inertial navigation," in *Proc. AAAI Conf. Artif. Intell.*, 2022, pp. 5413–5421.

[22] X. Cao, C. Zhou, D. Zeng, and Y. Wang, "Rio: Rotation-equivariance supervised learning of robust inertial odometry," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 6614–6623.

[23] Y. Wang, H. Cheng, A. Zhang, and M. Q.-H. Meng, "From IMU measurement sequence to velocity estimate sequence: An effective and efficient data-driven inertial odometry approach," *IEEE Sens. J.*, vol. 23, no. 15, pp. 17117–17126, Aug. 2023.

[24] J. Wang et al., "FedINS2: A federated-edge-learning-based inertial navigation system with segment fusion," *IEEE Internet Things J.*, vol. 11, no. 2, pp. 3653–3661, Jan. 2024.

[25] B. Zeinali, H. Zanddizari, and M. J. Chang, "IMUNet: Efficient regression architecture for inertial IMU navigation and positioning," *IEEE Trans. Instrum. Meas.*, vol. 73, 2024, Art. no. 2516213.

[26] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "MobileNetV2: Inverted residuals and linear bottlenecks," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 4510–4520.

[27] Y. Wang, H. Cheng, and M. Q.-H. Meng, "Pedestrian motion tracking by using inertial sensors on the smartphone," in *Proc. IEEE Int. Conf. Intell. Robots Syst.*, 2020, pp. 4426–4431.

[28] T. Feng, Y. Liu, Y. Yu, L. Chen, and R. Chen, "A data and physical model dual-driven based trajectory estimator for long-term navigation," *Def. Technol.*, vol. 40, pp. 78–90, 2024.

[29] Y. Wang, J. Xu, and Y. Zhao, "Wavelet encoding network for inertial signal enhancement via feature supervision," *IEEE Trans. Ind. Inform.*, vol. 20, no. 11, pp. 12924–12934, Nov. 2024.

[30] L. Yang et al., "Diffusion models: A comprehensive survey of methods and applications," *ACM Comput. Surv.*, vol. 56, no. 4, pp. 1–39, 2023.

[31] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, pp. 6840–6851.

[32] J. Song, C. Meng, and S. Ermon, "Denoising diffusion implicit models," in *Proc. Int. Conf. Learn. Representations*, 2021, pp. 1–20.

[33] Y. Song, J. Sohl-Dickstein, D. P. Kingma, A. Kumar, S. Ermon, and B. Poole, "Score-based generative modeling through stochastic differential equations," in *Proc. Int. Conf. Learn. Representations*, 2021, pp. 1–36.

[34] M. Kollovieh, A. F. Ansari, M. Bohlke-Schneider, J. Zschiegner, H. Wang, and Y. B. Wang, "Predict, refine, synthesize: Self-guiding diffusion models for probabilistic time series forecasting," in *Proc. Adv. Neural Inf. Process. Syst.*, 2023, pp. 28341–28364.

[35] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 10684–10695.

[36] Z. Kong, W. Ping, J. Huang, K. Zhao, and B. Catanzaro, "Diffwave: A versatile diffusion model for audio synthesis," in *Proc. Int. Conf. Learn. Representations*, 2021, pp. 1–17.

[37] Z. Li, S. Qi, Y. Li, and Z. Xu, "Revisiting long-term time series forecasting: An investigation on linear mapping," 2023, *arXiv:2305.10721*.

[38] J. Cui, Z. Tian, Z. Zhong, X. Qi, B. Yu, and H. Zhang, "Decoupled Kullback-Leibler divergence loss," in *Proc. Adv. Neural Inf. Process. Syst.*, 2024, pp. 74461–74486.

[39] G. Daras, M. Delbracio, H. Talebi, A. Dimakis, and P. Milanfar, "Soft diffusion: Score matching with general corruptions," *Trans. Mach. Learn. Res.*, 2023. [Online]. Available: https://openreview.net/forum?id=W98rebBxlQ

[40] K. Kim, T. Kwon, and J. C. Ye, "Noise distribution adaptive self-supervised image denoising using Tweedie distribution and score matching," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 2008–2016.

[41] V. De Bortoli, M. Hutchinson, P. Wirnsberger, and A. Doucet, "Target score matching," 2024, *arXiv:2402.08667*.

[42] Z. Yang, Y. Yu, C. You, J. Steinhardt, and Y. Ma, "Rethinking bias-variance trade-off for generalization of neural networks," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 10767–10777.

[43] H. Jiang et al., "Fast-DDPM: Fast denoising diffusion probabilistic models for medical image-to-image generation," *IEEE J. Biomed. Health Inform.*, vol. 29, no. 10, pp. 7326–7335, Oct. 2025.

Yingying Wang received the Ph.D. degree in electronic engineering from the Department of Electronic Engineering, The Chinese University of Hong Kong, Hong Kong, in 2023.

She was a Postdoctoral Researcher with Hong Kong Automotive Platforms and Application Systems (APAS) R&D center, Hong Kong, from 2023 to 2024. She is currently a Postdoctoral Fellow with the Trustworthy AI and Autonomous Systems Lab, Hong Kong Polytechnic University, Hong Kong. Her research interests include pedestrian localization and nonintrusive intelligent sensing.

Shaoting Qiu received the dual bachelor's degrees in computer engineering and business and law from the Beijing University of Posts and Telecommunications, Beijing, China, in 2024. She is currently working toward the Ph.D. degree in aeronautical and aviation engineering with the Department of Aeronautical and Aviation Engineering, The Hong Kong Polytechnic University, Hong Kong.

Her research interests include SLAM, sensor fusion, and localization for autonomous systems.

Yulong Huang (Senior Member, IEEE) received the B.S. degree in automation and the Ph.D. degree in control science and engineering from the College of Automation, Harbin Engineering University (HEU), Harbin, China, in 2012 and 2018, respectively.

From 2016 to 2017, he was a Visiting Graduate Researcher with the Electrical Engineering Department, Columbia University, New York, USA. From 2019 to 2021, he was associated with the Department of Mechanical Engineering, City University of Hong Kong, Hong Kong, as a Hong Kong Scholar. He is currently a Full Professor of Navigation, Guidance, and Control with HEU. His research interests include state estimation, intelligent information fusion and their applications in navigation technology, such as inertial navigation, integrated navigation, intelligent navigation, and cooperative navigation.

Dr. Huang serves as an Associate Editor for IEEE TRANSACTIONS ON AUTOMATIC CONTROL, IEEE TRANSACTIONS ON AEROSPACE AND ELECTRONIC SYSTEMS, *IEEE Aerospace and Electronic Systems Magazine*, IEEE TRANSACTIONS ON AUTOMATION SCIENCE AND ENGINEERING, IEEE TRANSACTIONS ON INSTRUMENTATION AND MEASUREMENT, and IEEE SENSORS JOURNAL, and a Youth Editor for the IEEE/CAA JOURNAL OF AUTOMATICA SINICA, and *Satellite Navigation*.

Weisong Wen (Member, IEEE) received the B.Eng. degree in mechanical engineering from Beijing Information Science and Technology University (BISTU), Beijing, China, in 2015, the M.Eng. degree in mechanical engineering from the China Agricultural University, Beijing, China, in 2017, and the Ph.D. degree in mechanical engineering from The Hong Kong Polytechnic University (PolyU), Hong Kong, in 2020.

He was also a visiting Ph.D. student with the Faculty of Engineering, University of California, Berkeley, Berkeley, CA, USA, in 2018. Since 2021, he has been a Research Assistant Professor with Department of Aeronautical and Aviation Engineering, PolyU, where he joined as an Assistant Professor, in 2023. His research interests include trustworthy multisensory integration, LiDAR SLAM, GNSS positioning and autonomous systems.